

PUBLISHED

Question	Answer	Marks	Guidance
5	$H_0: \mu_A = \mu_B \quad H_1: \mu_A \neq \mu_B$	B1	If in words, must have population means. Allow $H_0: \mu_d = 0$, $H_1: \mu_d \neq 0$.
	'1331848' – $\frac{'3264'^2}{8}$ [=136]	M1	OE, may be implied by $s_A^2 = 19.43$. Biased estimate of 17 earns M0 unless recovered.
	'964964' – $\frac{'2406'^2}{6}$ [=158]	M1	OE, may be implied by $s_B^2 = 31.6$. Biased estimate of 26 earns M0 unless recovered.
	$s_p^2 = \frac{'136' + '158'}{8 + 6 - 2} = 24.5$	M1	Correct form using <i>their</i> values, may be embedded.
		A1	OE, for example $\frac{49}{2}$.
	$t = \frac{\frac{'3264'}{8} - \frac{'2406'}{6}}{'s_p' \sqrt{\frac{1}{8} + \frac{1}{6}}} = 2.62$	M1	Correct form using <i>their</i> values.
		A1	AWRT 2.62.
	'2.62' > 2.179, reject H_0 / significant.	M1	Compare <i>their</i> 2.62 with 2.179 and appropriate result (may be in terms of H_1). Test result may be implied by an attempt at an appropriate conclusion in context ignoring hypotheses.
	Sufficient evidence to suggest that there is a difference in the mean tensile strengths of steel rods produced by the two machines.	A1	Correct conclusion from correct working ignoring hypotheses. In context. Level of uncertainty in language is used (for example, not 'prove'). Allow "(Some/enough) evidence that / to show / conclude that...". Do not accept statements such as "there is insufficient evidence to suggest...".
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